# Product Description

| Product Information | |
|---|---|
| Product name | Low-Noise Wideband Amplifier |
| Product number | LNA-03-30-A-R |
| Manufacturer | FLIM LABS S.r.l. |
| Registered Address | Via della Farnesina, 3 – 00135 Rome (RM), Italy |




## General Description

The low-noise broadband amplifier is a compact electronic module designed for amplification of high-frequency electronic signals, in particular low-amplitude pulsed signals in the millivolt range used in scientific instrumentation, photon counting, and fluorescence lifetime imaging microscopy (FLIM) applications.

The device receives electronic input signals from external devices such as single-photon detectors, discrimination modules, or other front-end electronics, and provides an amplified output signal compatible with downstream acquisition, timing, and analysis instruments. Input and output signals are connected via standard SMA coaxial connectors suitable for high-speed, low-distortion signal transmission.

The module features:

- Compact electronic amplifier module
- One SMA input for the signal to be amplified
- One SMA output for the amplified signal
- Non-inverting amplification
- Broadband operation up to the specified cut-off frequency
- Low timing variation
- Compact aluminum enclosure suitable for system integration
- USB Type-C power supply
- Power status LED

# Intended Use and Misuse

## Intended Use

The device is intended as a compact electronic module for amplification of electronic signals by professional users in research laboratories, universities, or industrial environments. It may be used as a stand-alone unit or integrated into custom measurement systems.

## Reasonably Foreseeable Misuse

The following uses are not permitted:

- Use outside described applications
- Use with incompatible components
- Operation outside specified limits
- Unauthorized modification
- Use with incorrect power supply

# Environmental and Operating Conditions

The device is designed for indoor laboratory use only. Operate the device only within the specified environmental limits:

- Operating temperature: 0°C to +40°C
- Storage temperature: −20°C to +70°C
- Relative humidity: up to 80% non-condensing
- Stable, vibration-free surface

Do not operate the device in explosive atmospheres or in the presence of flammable gases. Ensure adequate ambient lighting during installation and adjustment to avoid connection errors.